

Yunus Emre Danabaş

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EDUCATION

Sabanci University, *Mechatronic Engineering / Electronics Engineering (Double Major)* 2020 – 2025
Istanbul, Turkey

- CGPA 3.77 (Dean's High Honor) | Ranked 3rd among 33 seniors.
- Honors: Full Scholarship | 2023 and 2024 Extracurricular Student Activities Leadership Award
- Related Courses: Kinematics and Dynamics of Machines - Autonomous Mobile Robotics - Introduction to Robotics - Deep Learning for Robot Control.

WORK EXPERIENCE

Sabanci University, *Teaching Assistant for Prof. Volkan Patoğlu* Sep 2024 – present

- Assisted in "ME312 Analysis and Synthesis of Mechanisms" and "ME403 Introduction to Robotics" instructing 20+ students, holding weekly office hours, and supervising course projects.

Munich Institute of Robotics and Machine Intelligence, Technical University of Munich (TUM - MIRMI), *Research Intern, Supervised by M.Sc. Mehmet Can Yıldırım* Jun 2024 – Aug 2024
Munich, Germany

- Investigated force and moment scaling in robotic force-torque sensors, focusing on optimizing sensor design for higher sensitivity and load capacity.
- Designed and fabricated flexure hinge prototypes using MATLAB for simulation and SolidWorks for CAD modeling, integrating strain gauge systems for deformation measurement.
- Conducted experimental tests on flexure hinges, utilizing computer vision techniques for real-time analysis and validation of deformation models.
- Developed digital twins in Gazebo, performing preliminary simulation experiments for robotic system stability and dynamic response assessment.

TUBITAK BILGEM, *Robotics Engineering Intern* Sep 2023 – Oct 2023
Kocaeli, Turkey

- Enhanced skills in ROS through formal training, focusing on robotic simulations.
- Conducted comparative analyses of single and multi-robot simulations with Gazebo and RViz, exploring frontier-based exploration and map merging.

Pubinno Inc., *Mechanical Engineering Intern* Jun 2023 – Sep 2023
Istanbul, Turkey

- Designed 3D-printed prototypes using Onshape, expediting the production process.
- Led the development of an automated maintenance prototype for field staff to perform initial maintenance.
- Coordinated quality control activities, collaborating with manufacturers and documenting QA/QC procedures.

TECHNICAL SKILLS / LANGUAGES

- CAD & 3D Printing:** OnShape, SolidWorks (CSWA Certified), 3D printing (machine operation, calibration, part replacement, design optimization)
- Robotics & Electronics:** Robot Operating System (ROS), Gazebo, MuJoCo, Arduino, ESP32, LTSpice
- Software & Programming:** Python (JAX, Equinox), C++, MATLAB, Simulink, Linux, LaTeX, Autolev (MotionGenesis), Verilog
- Languages:** English (Advanced), Turkish (Native)

PROJECTS

Data-Driven Control of Exoskeletons, Oct 2024 – present
Graduation Project | Supervisors: Prof. Volkan Patoğlu, Dr. Aykut Cihan Satıcı

- Developing a control system for exoskeletons using data-driven optimization and human-in-the-loop (HiL) learning to dynamically adapt to individual users and varying environments.
- Conducting simulations in MuJoCo to design and test advanced control strategies, focusing on improving energy efficiency, user comfort, and adaptability.
- Implementing Bayesian optimization techniques with Gaussian Processes for real-time feedback adjustment.
- Transitioning from simulation to prototype testing for experimental validation in assistive robotics.

How can a robot place an item on a cluttered desk?,

Supervised by Professors Volkan Patoğlu and Esra Erdem

Jan 2023 –

Jan 2025

- Designed and implemented an intelligent approach for robotic item placement on cluttered surfaces using Gazebo simulations and the Baxter robot.
- Optimized ROS Noetic compatibility for the Baxter robot, resolving MoveIt integration challenges for simulated and real-world applications.

SURover (Sabancı University Rover Team), *Mechanical Team Member*

Jun 2022 –

Jan 2025

- Conducted ROS-based simulations for robot arm control with MOVEIT and tested joystick and sensor functions on TurtleBots in Gazebo to refine simulation environments.
- Performed ArUco tag detection and fiducial SLAM using the Zed2 stereo camera, enhancing localization accuracy.
- Utilized SolidWorks for mechanical design, focusing on efficient, optimized solutions, and proficiently implemented motor control using Arduino across various applications.

TRAININGS

Autonomous Driving Technologies Specialization Program, *National Technology Academy*

Sep 2024 –

Jun 2025

- Comprehensive training in autonomous driving technologies, including sensor fusion, vehicle modeling, ADAS, and motion planning, with practical applications in deep learning and environment perception.

CAMPus – Learning & Development Camp for Engineers, *ROKETSAN*

Sep 2023 –

Nov 2023

- Completed technical modules on rocket propulsion systems (solid, liquid, RAMJET, SCRAMJET) and specialized chemical material properties training, enhancing aerospace technology and engineering fundamentals expertise.

ACTIVITIES

Sabancı University SIAM (Society for Industrial and Applied Mathematics) Student Chapter,

Financial Affairs Coordinator

Sep 2022 –

Jun 2024

- Managed financial matters, budgets, and communications with global SIAM authorities.
- Coordinated seminars, reading groups, and mathematics competitions.

Civic Involvement Projects, *Volunteer*

Apr 2021 –

Jun 2021

- Narrated audiobooks for visually impaired individuals, contributing to accessibility and community support.